

GENERAL CHEMISTRY



Chapter 20 Electrochemistry

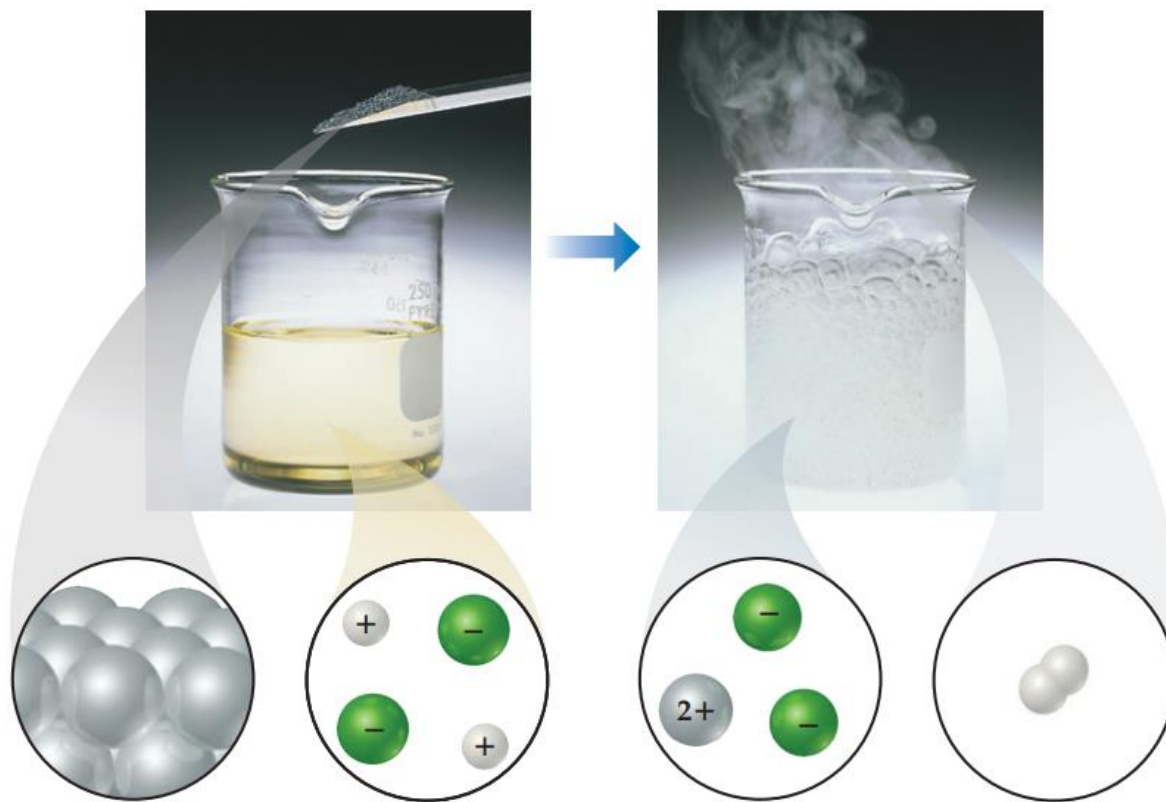
Expected Learning Outcomes

- ◆ Determine atom's individual oxidation number
- ◆ Determine which element is oxidized and which element is reduced in a reaction
- ◆ Balance redox reaction using the half-reaction approach
- ◆ Know how voltaic (galvanic) cells work
- ◆ Know how electrolytic cell and batteries work

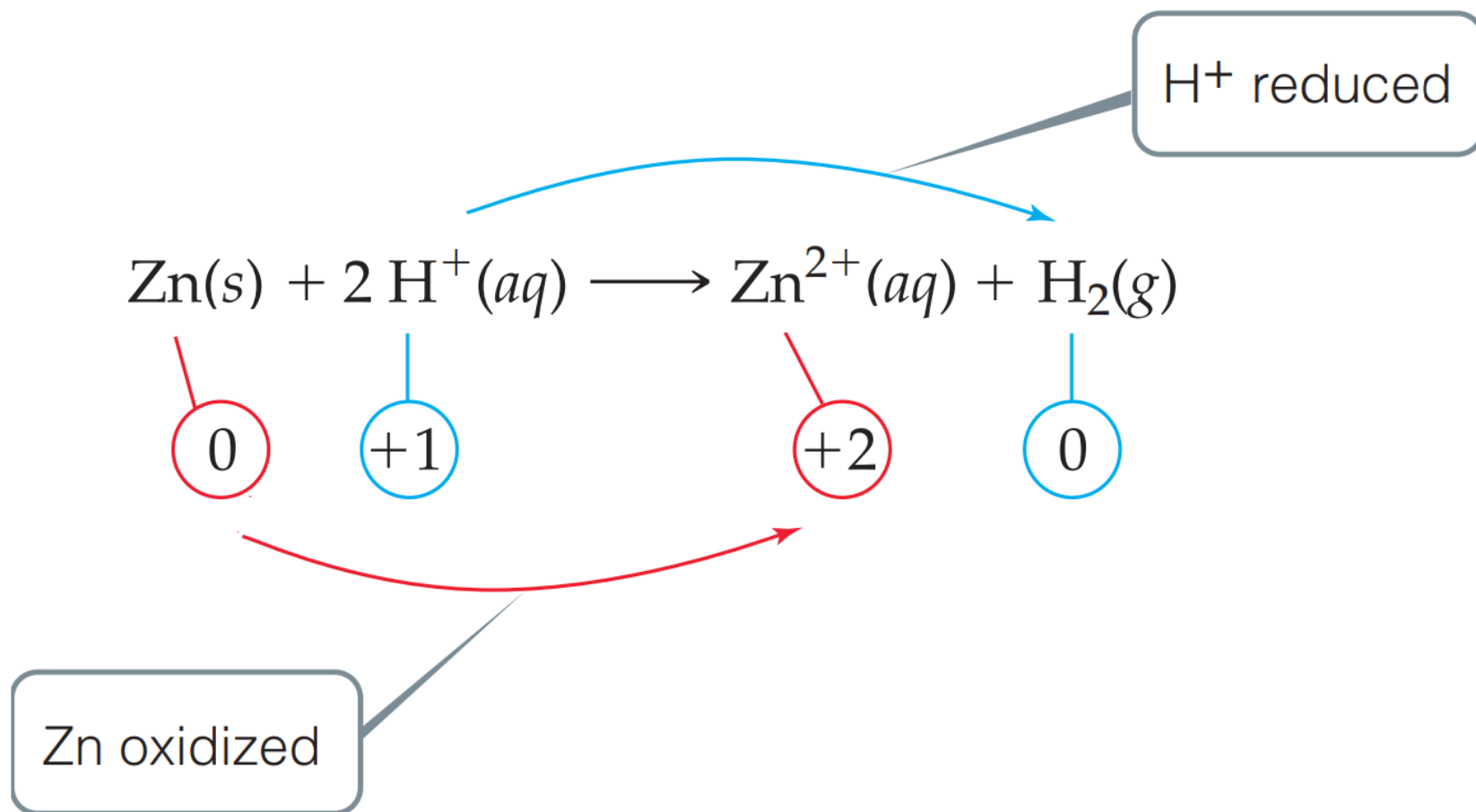
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- 20-7 Batteries and Fuel Cells
- 20-8 Corrosion (*skipped*)
- 20-9 Electrolysis (*skipped*)

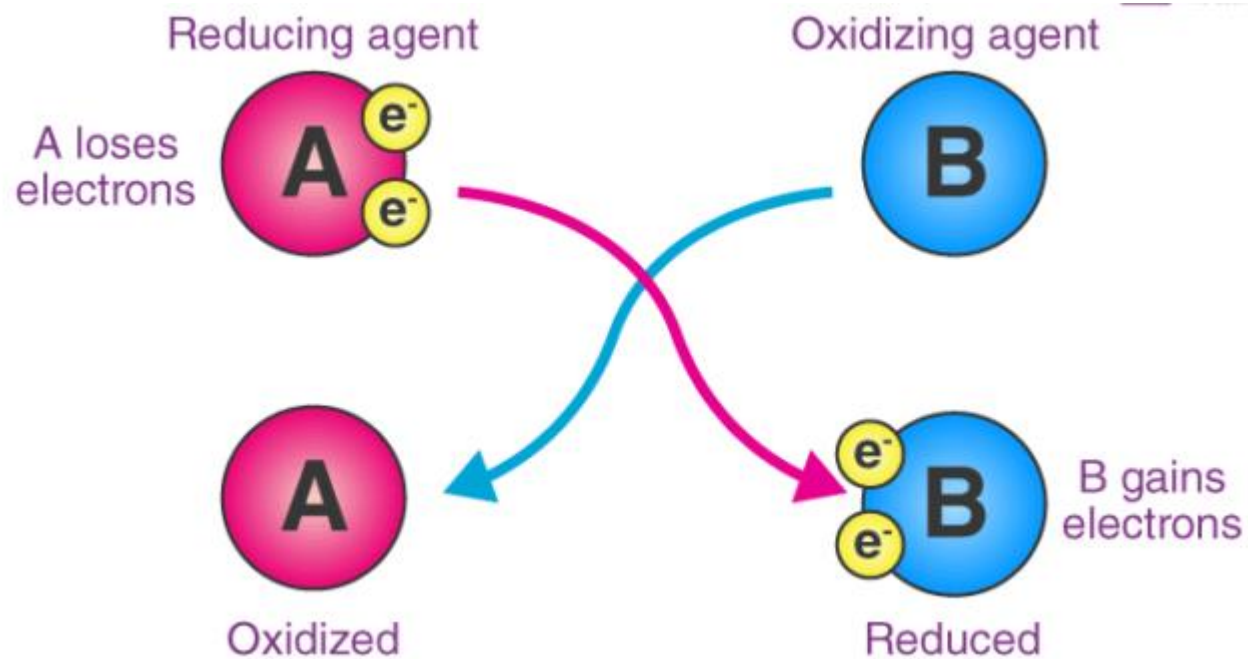
20-1 Oxidation States and Oxidation–Reduction Reactions



Redox reaction



Redox reaction



Rule for assigning oxidation number

General rule:

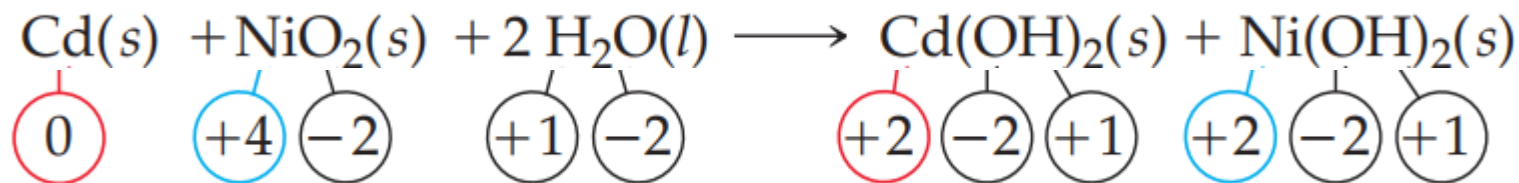
1. For an atom in its elemental form (Na, O₂, S₈, etc.): **O.N. = 0**
2. For an ion the O.N. equals the charge: Na⁺, Ca²⁺, NO₃⁻

Specific rule:

3. For hydrogen:
O.N. = +1 with nonmetals **Ex: HCl, HClO, H₂O**
O.N. = -1 with metals and boron **Ex: NaH, BrH₃**
4. For Fluorine: O.N. = -1
5. For oxygen:
O.N. = -1 in peroxides (X₂O₂, X = Group 1(A) element) **Ex: Na₂O₂**
O.N. = $-\frac{1}{2}$ in superoxides (XO₂, X = Group 1(A) element) **Ex: Na₂O**
O.N. = -2 in all other compounds
6. Group 7A O.N. = -1 (except when connected to O)

EXAMPLE

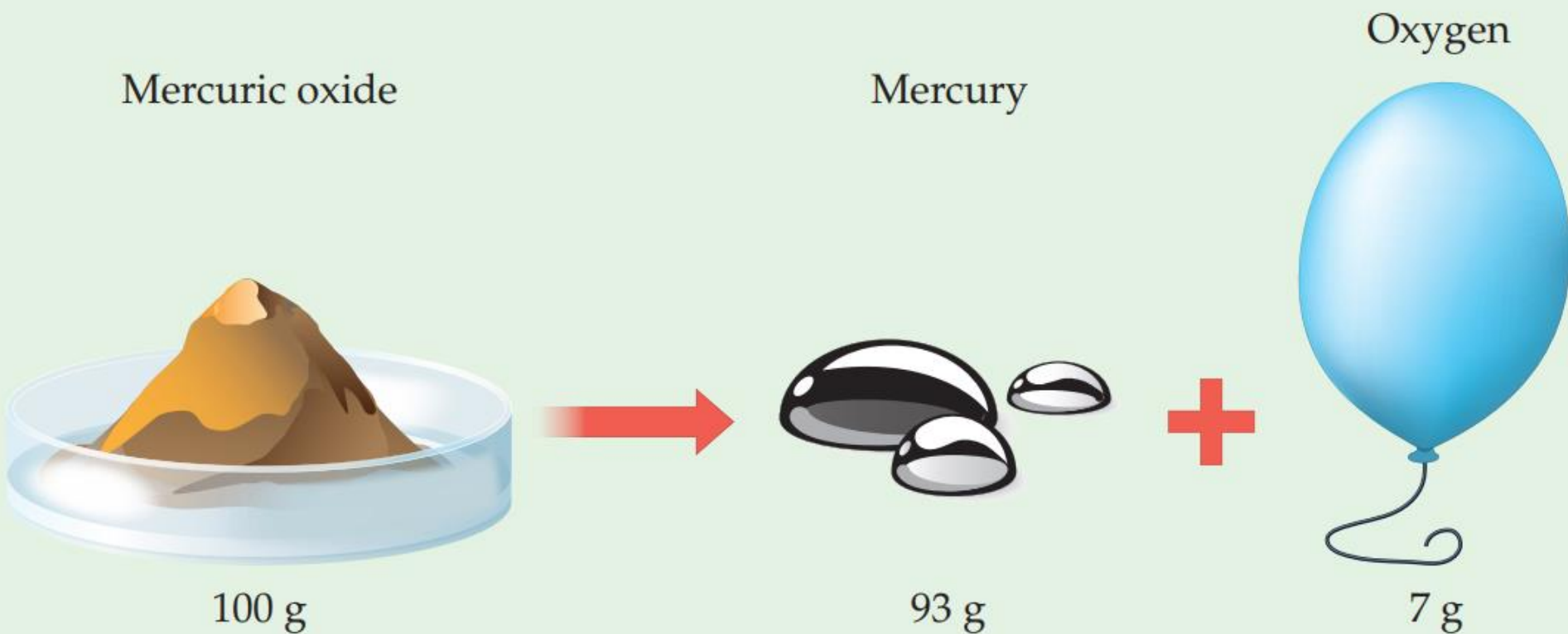
The nickel–cadmium (nicad) battery uses the following redox reaction to generate electricity. Identify the substances that are oxidized and reduced, and indicate which is the oxidizing agent and which is the reducing agent.



Cd atom is oxidized (loses electrons) and is the reducing agent

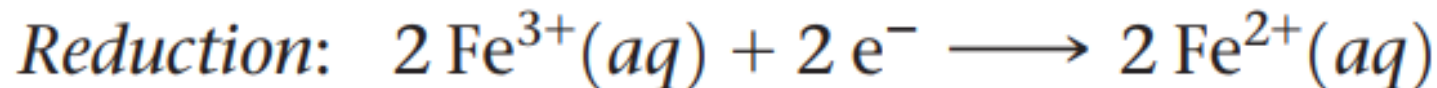
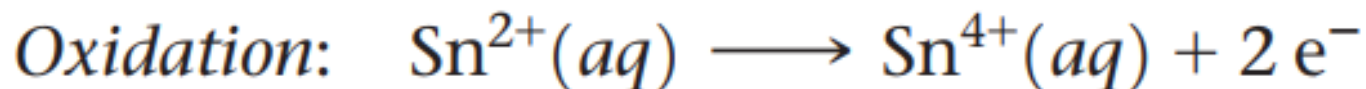
NiO_2 is reduced (gains electrons) and is the oxidizing agent

20-2 Balancing redox reactions

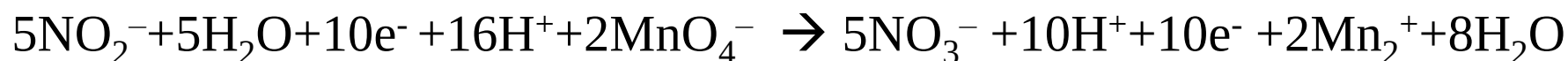
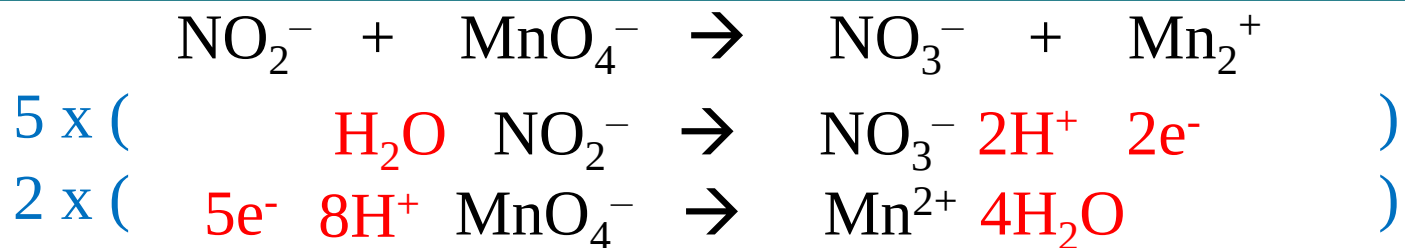


Law of Conservation of Mass

Half reaction



Balance Redox Reactions in Acidic Aqueous Solution



STEP 1: Write the equation into 2 half-reactions.

STEP 2: Balance elements that are not oxygen or hydrogen.

STEP 3: Balance Oxygens by adding H₂O.

STEP 4: Balance Hydrogens by adding H⁺.

STEP 5: Balance overall charge by adding electrons (e⁻) to the more **positive** side. → **Electrons from both half-reactions must match**

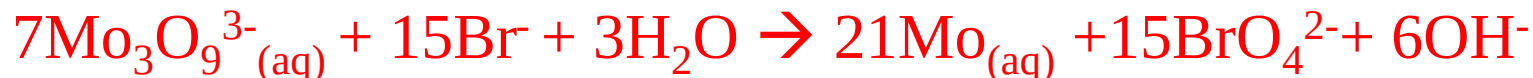
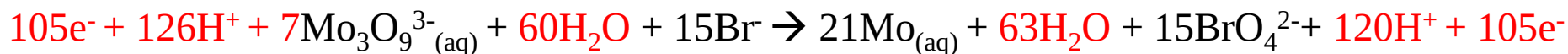
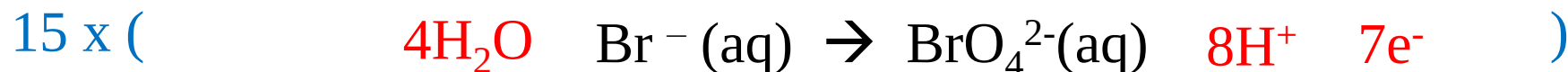
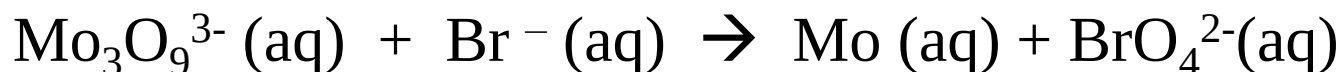
STEP 6: Combine the half-reactions and cross out reaction intermediates.

Balance Redox Reactions in Basic Aqueous Solution

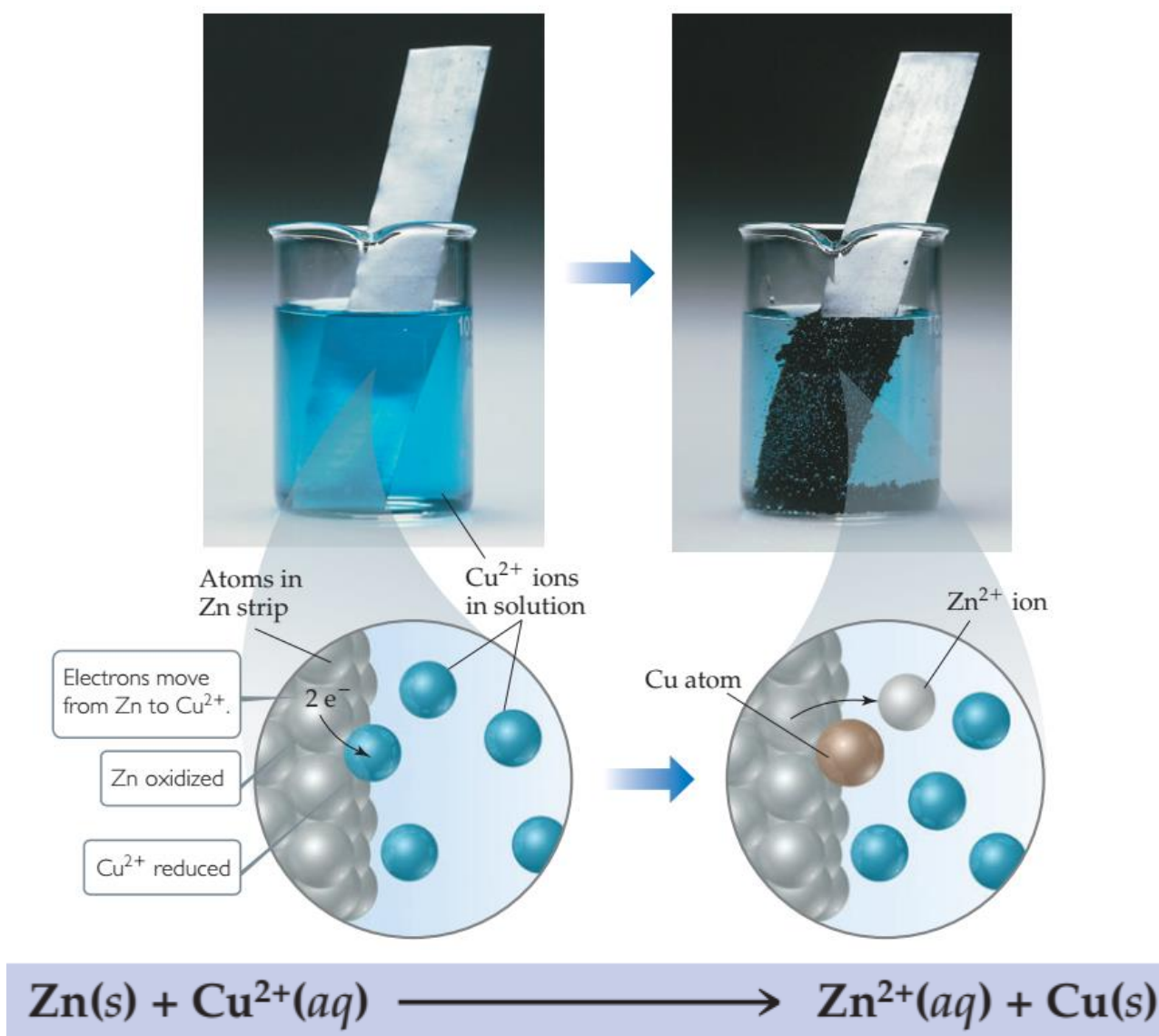
Follow Steps 1-6 from above.

STEP 7: Balance H^+ by adding OH^- ions to both sides of the chemical reaction.

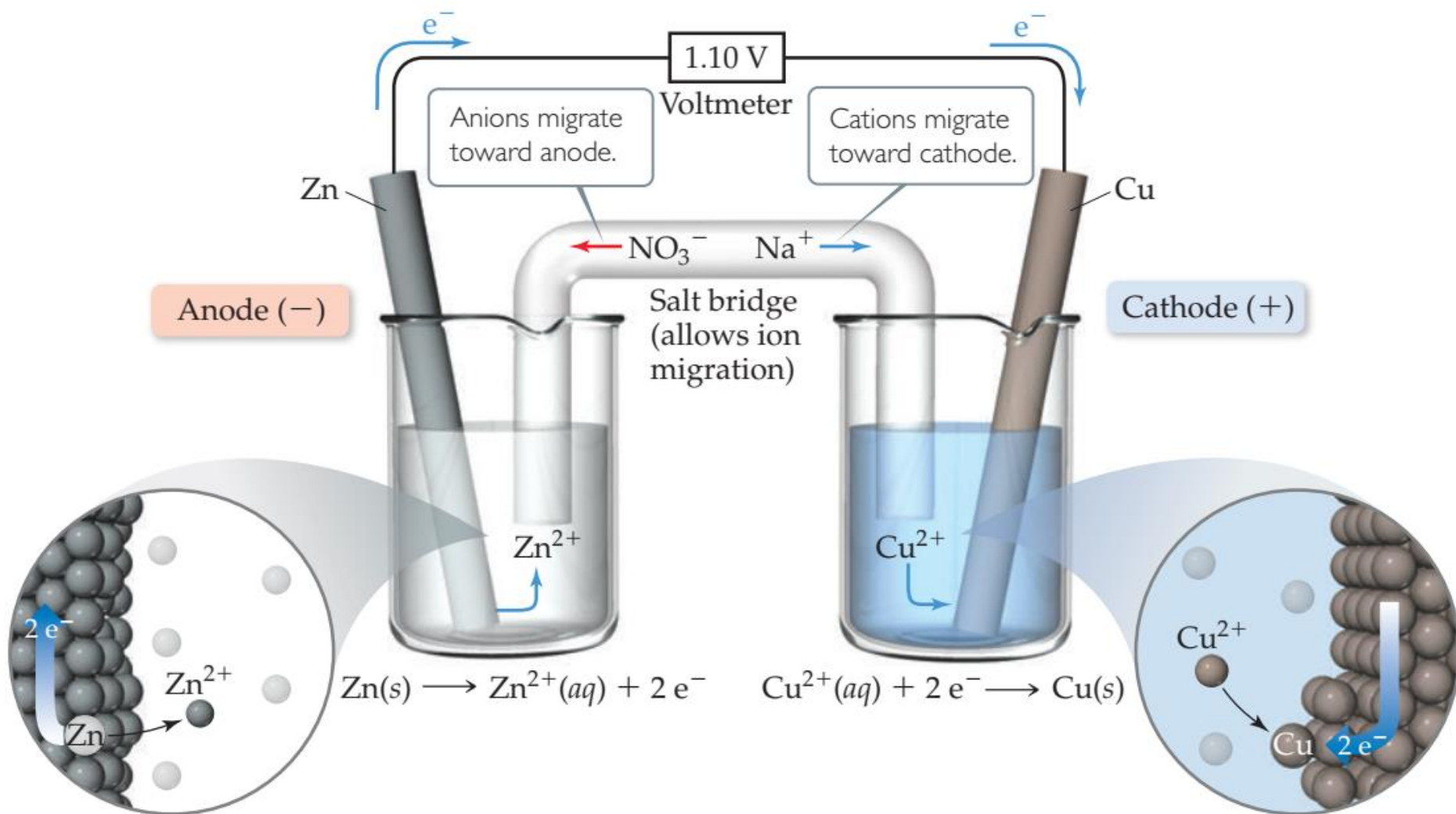
Example 2: Balance the following reaction in a basic solution.



20-3 Voltaic cell



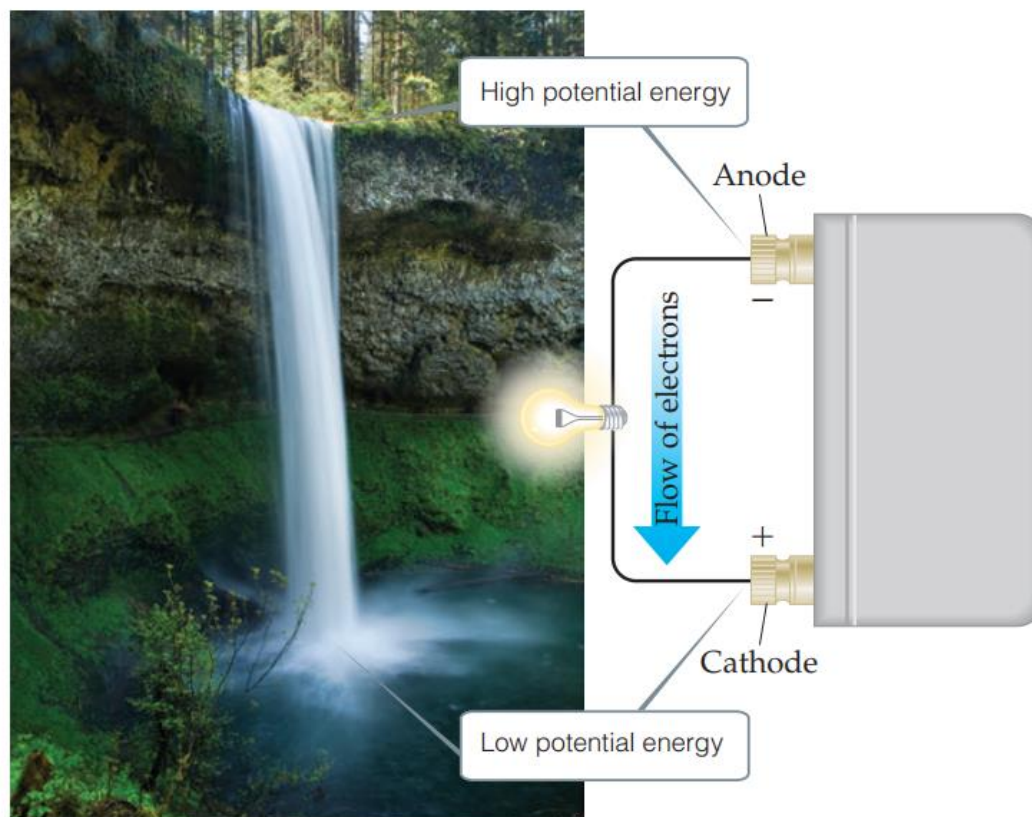
Cu – Zn voltaic cell



***Anions always migrate toward the anode
and cations toward the cathode***

Q: Why do electrons transfer spontaneously from a Zn atom to a Cu^{2+} ion?

A: It's because of a difference in potential energy

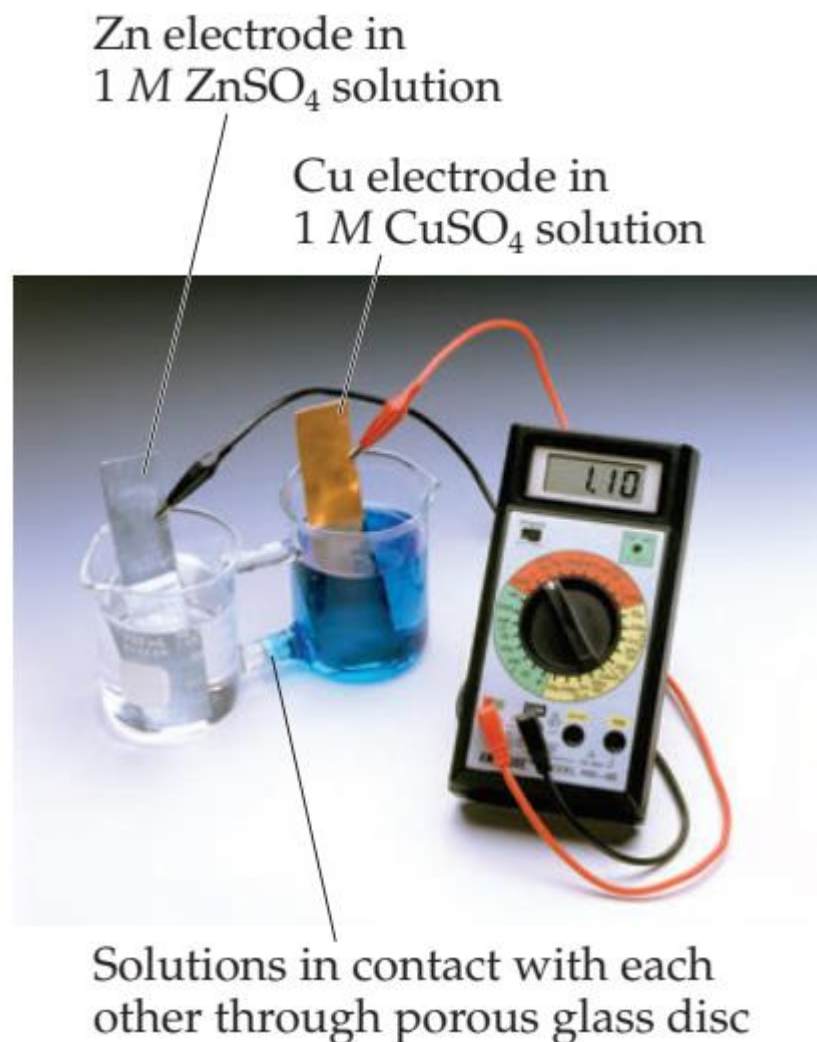


The potential difference between the two electrodes of a voltaic cell is called the **cell potential** (E_{cell}) or **electromotive force**, or **emf**

20-4 Cell potential under standard conditions

Standard states:

State of Matter	Standard State
Solid	Pure solid
Liquid	Pure liquid
Gas	100.0 kPa pressure
Solution	1 M concentration
Element	$\Delta G_f^\circ = 0$ for element in standard state



Standard reduction potential

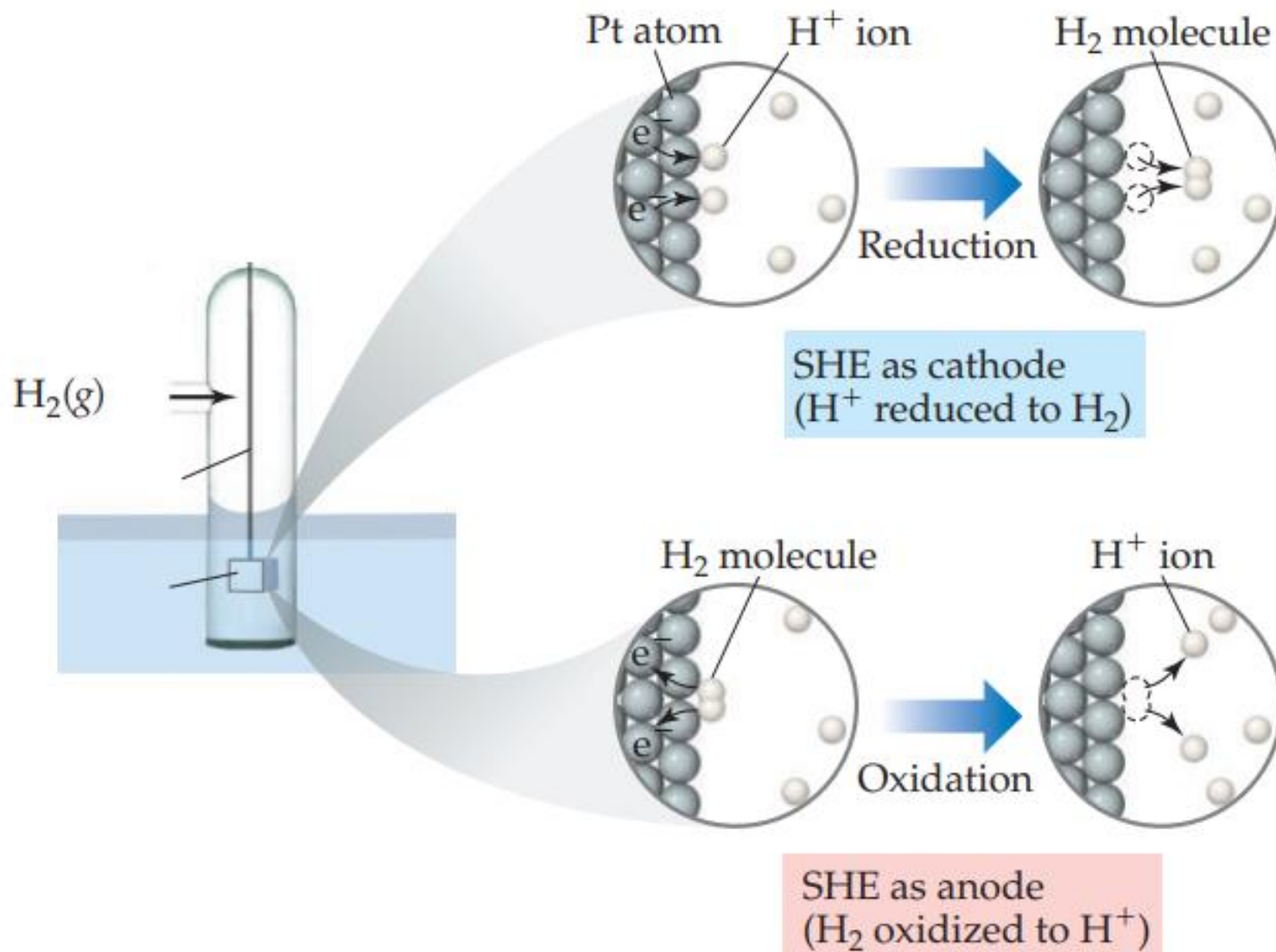
$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ} (\text{cathode}) - E_{\text{red}}^{\circ} (\text{anode})$$

It is not possible to measure the standard reduction potential of a half-reaction directly

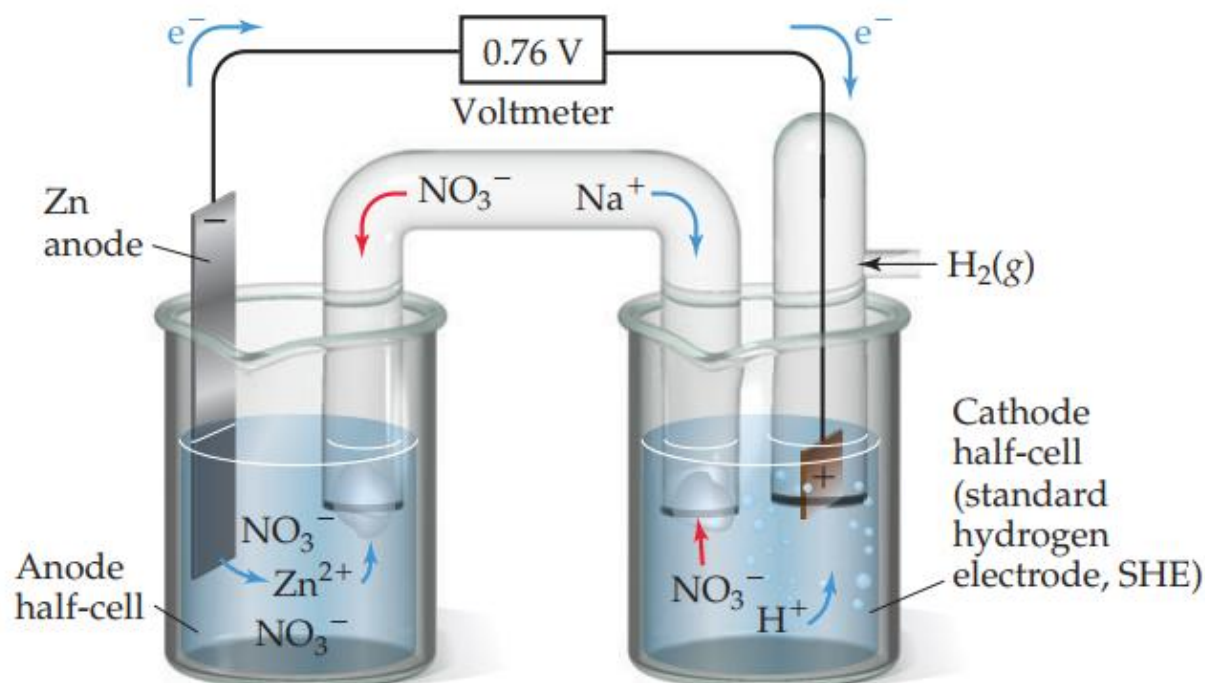
➔ Need a reference half-reaction



Standard hydrogen electrode (SHE) as a reference electrode



Standard hydrogen electrode (SHE) as a reference electrode



$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ}(\text{cathode}) - E_{\text{red}}^{\circ}(\text{anode})$$

$$+0.76 \text{ V} = 0 \text{ V} - E_{\text{red}}^{\circ}(\text{anode})$$

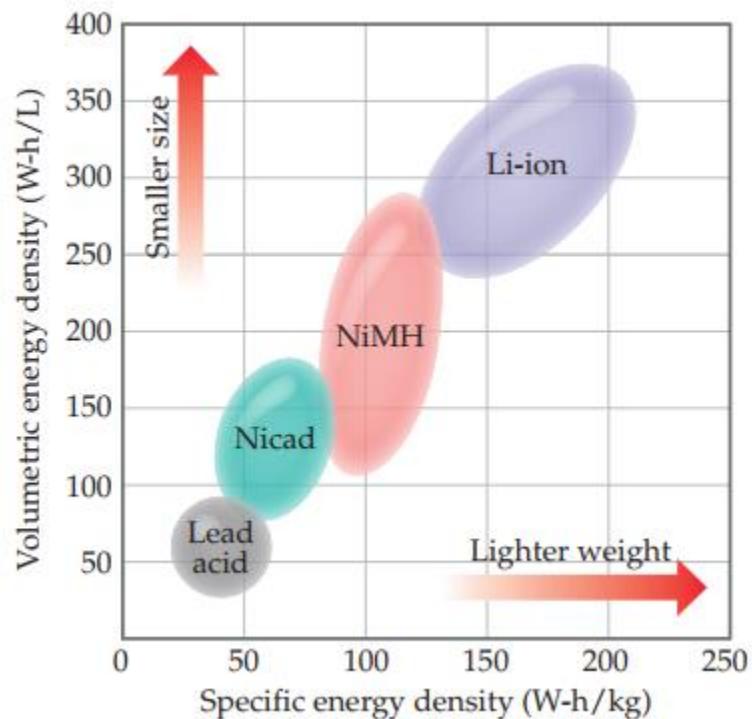
$$E_{\text{red}}^{\circ}(\text{anode}) = -0.76 \text{ V}$$



Intensive property



20-5 Free energy and redox reaction



- Electric vehicles have engine efficiencies of $\sim 70\%$, while petrol and diesel vehicle have engine efficiencies of 15-20%
- Since 2010, the cost of batteries has dropped by 600%

Relationship between E° , G° , and K

In general: $\Delta G = -nFE$

n : number of moles of electrons transferred

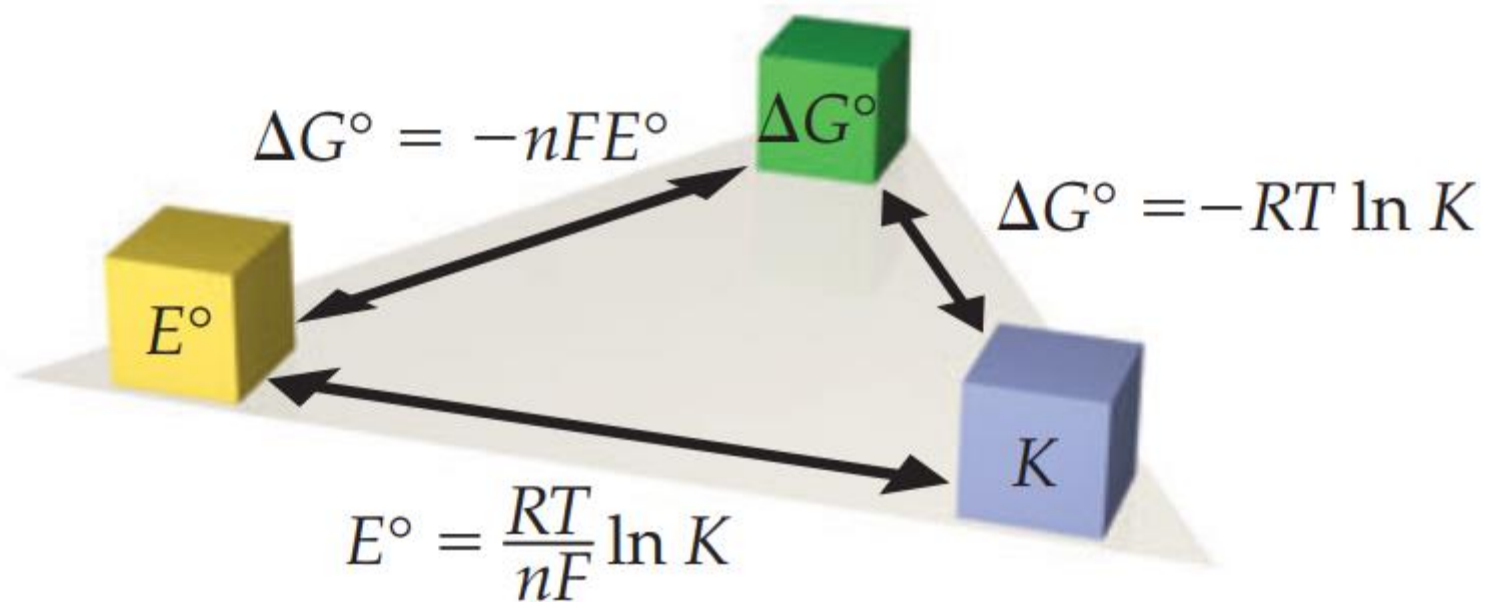
F : Faraday constant: quantity of electrical charge on 1 mol of electrons

$$F = 96,485 \text{ C/mol} = 96,483 \text{ J/Vmol}$$

When the reactants and products are all in their standard states

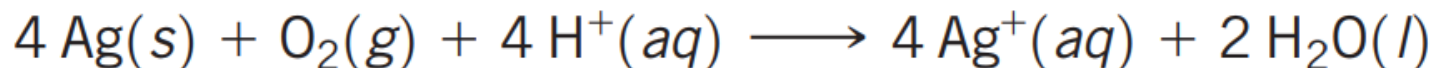
$$\Delta G^\circ = -nFE^\circ$$

$$E^\circ = \frac{\Delta G^\circ}{-nF} = \frac{-RT \ln K}{-nF} = \frac{RT}{nF} \ln K$$



EXAMPLE

(a) Calculate the standard free energy change, ΔG° , and the equilibrium constant, K , at 298 K for the reaction



$E^\circ_{\text{red}}(\text{V})$	Reduction Half-Reaction
+1.23	$\text{O}_2(g) + 4 \text{H}^+(aq) + 4 \text{e}^- \longrightarrow 2 \text{H}_2\text{O}(l)$
+0.80	$\text{Ag}^+(aq) + \text{e}^- \longrightarrow \text{Ag}(s)$



$$E^\circ = (1.23 \text{ V}) - (0.80 \text{ V}) = 0.43 \text{ V}$$

$$\Delta G^\circ = -nFE^\circ$$

$$= -(4)(96,485 \text{ J/V mol})(+0.43 \text{ V}) = -170 \text{ kJ/mol}$$

$$\Delta G^\circ = -RT \ln K$$

$$K = e^{-\Delta G^\circ/RT} = 9 \times 10^{29}$$

Cell potential under nonstandard conditions

Under nonstandard conditions:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

The relationship between emf and the free-energy change

$$\Delta G = -nFE$$

Therefore,
$$-nFE = -nFE^\circ + RT \ln Q$$

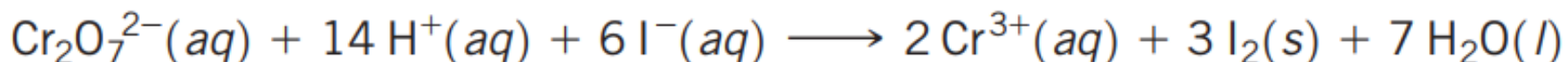
$$E = E^\circ - \frac{RT}{nF} \ln Q \quad \text{Nernst equation}$$

$$E = E^\circ - \frac{2.303 RT}{nF} \log Q$$

$$E = E^\circ - \frac{0.0592 \text{ V}}{n} \log Q \quad (T = 298 \text{ K})$$

EXAMPLE

Calculate the emf at 298 K generated by a voltaic cell in which the reaction is



When

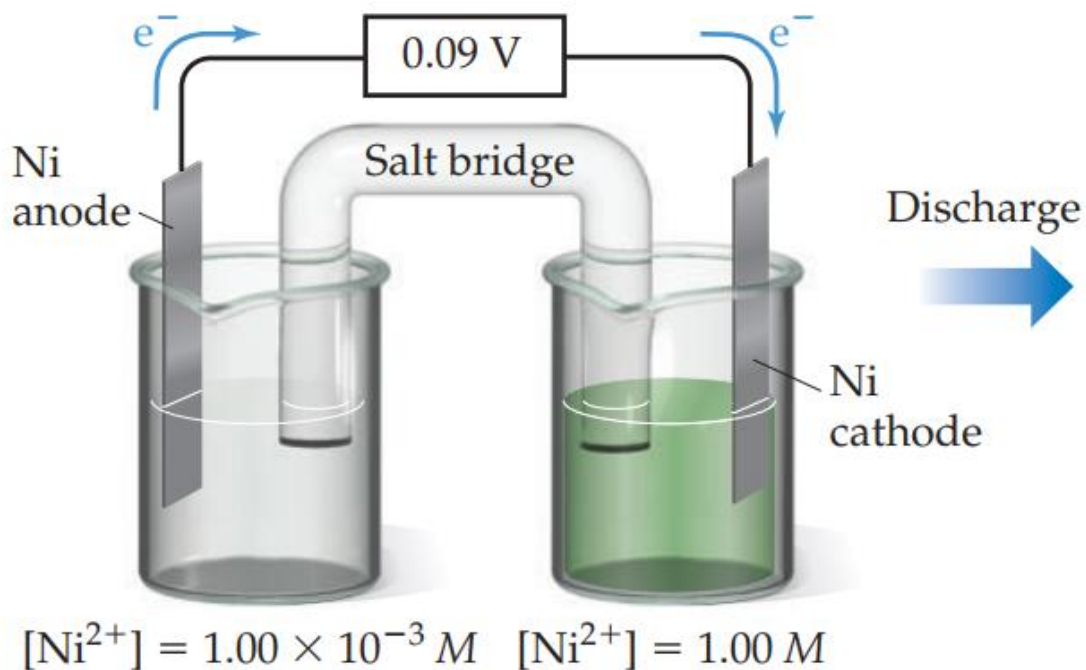
$$[\text{Cr}_2\text{O}_7^{2-}] = 2.0 \text{ M}, [\text{H}^+] = 1.0 \text{ M}, [\text{I}^-] = 1.0 \text{ M}, \text{ and } [\text{Cr}^{3+}] = 1.0 \times 10^{-5} \text{ M}$$

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

$$Q = \frac{[\text{Cr}^{3+}]^2}{[\text{Cr}_2\text{O}_7^{2-}][\text{H}^+]^{14}[\text{I}^-]^6} = \frac{(1.0 \times 10^{-5})^2}{(2.0)(1.0)^{14}(1.0)^6} = 5.0 \times 10^{-11}$$

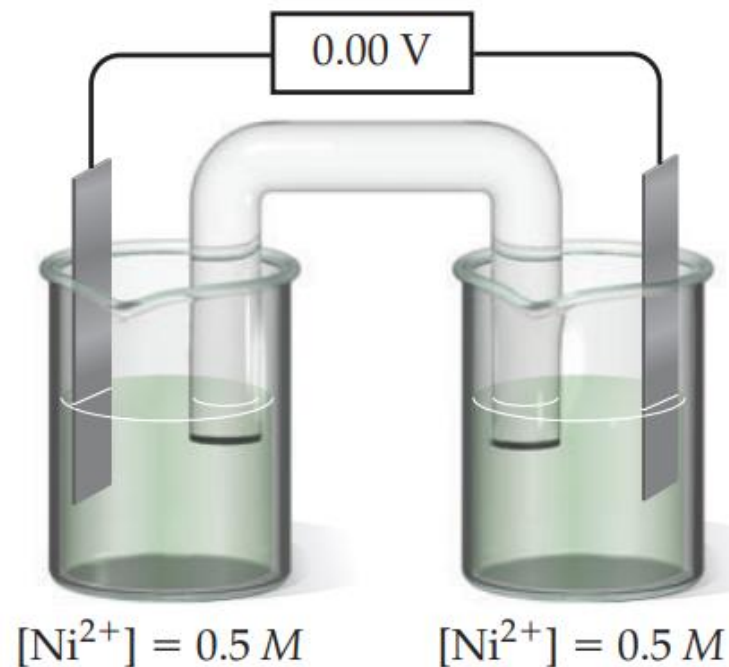
$$E = 0.79 \text{ V} - \left(\frac{0.0592 \text{ V}}{6} \right) \log(5.0 \times 10^{-11}) = 0.89 \text{ V}$$

Concentration cells



(a)

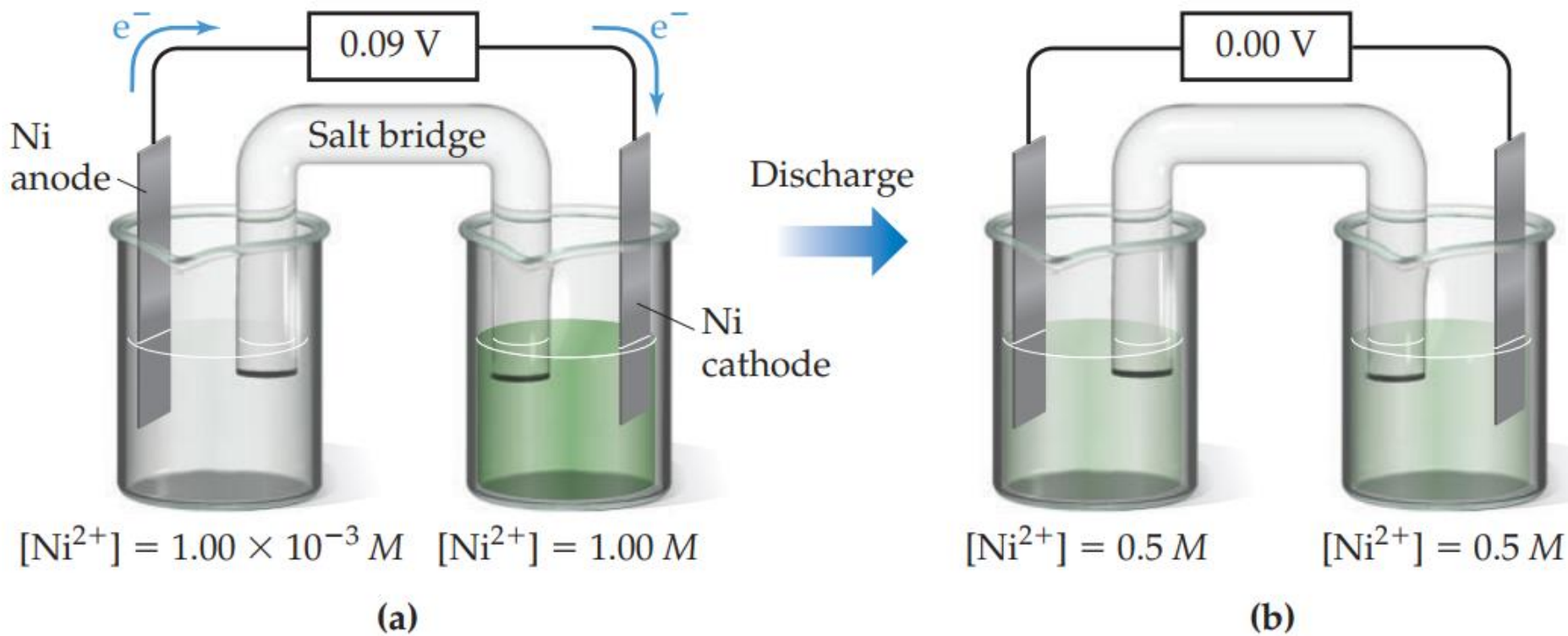
Concentrations of $\text{Ni}^{2+} (\text{aq})$ in the two half-cells are unequal,
→ the cell generates an electrical current and a voltage



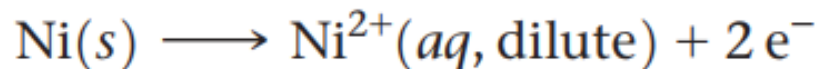
(b)

the cell has reached equilibrium and the emf goes to zero

Concentration cells



Anode:



Cathode:



Overall:



EXAMPLE

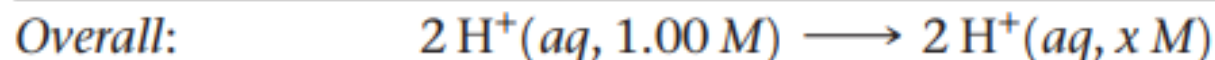
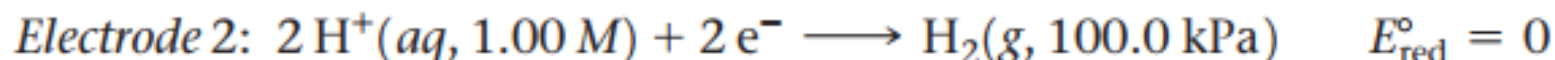
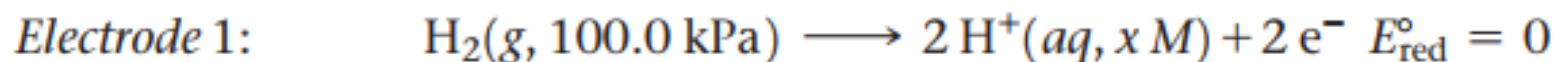
A voltaic cell is constructed with two hydrogen electrodes.

Electrode 1 has $P_{\text{H}_2} = 100.0 \text{ kPa}$ and an unknown concentration of $\text{H}^+ (aq)$. **Electrode 2** is a standard hydrogen electrode ($P_{\text{H}_2} = 100.0 \text{ kPa}$, $[\text{H}^+] = 1.00 \text{ M}$). At 298 K the measured cell potential is 0.211 V, and the electrical current is observed to flow from electrode 1 through the external circuit to electrode 2. What is the pH of the solution at electrode 1?

Nernst equation
$$E = E^\circ - \frac{2.303 RT}{nF} \log Q$$

$$0.211 \text{ V} = 0 - \frac{0.0592 \text{ V}}{2} \log Q$$

$$Q = 10^{-7.13} = 7.4 \times 10^{-8}$$



$$Q = \frac{[\text{H}^+(\text{aq}, x \text{ M})]^2}{[\text{H}^+(\text{aq}, 1.00 \text{ M})]^2} = \frac{x^2}{(1.00)^2} = x^2 = 7.4 \times 10^{-8}$$

$$x = [\text{H}^+] = \sqrt{7.4 \times 10^{-8}} = 2.7 \times 10^{-4}$$

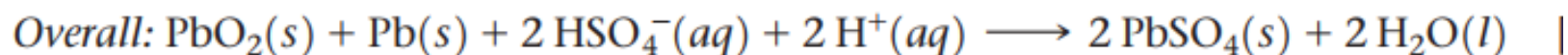
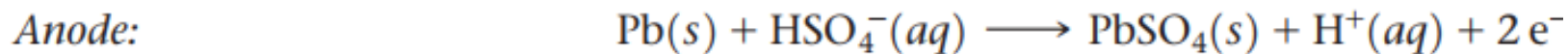
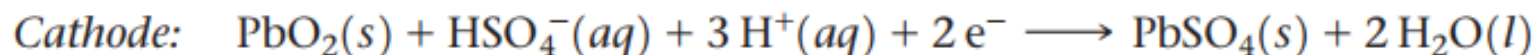
$$\text{pH} = -\log[\text{H}^+] = -\log(2.7 \times 10^{-4}) = 3.57$$

20-7 Batteries and Fuel Cells



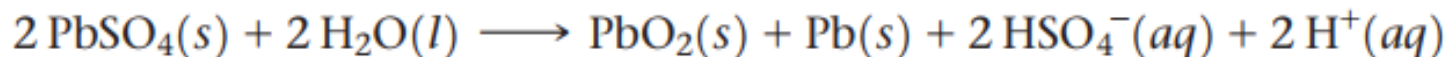
A **battery** is a portable, self-contained electrochemical power source that consists of one or more voltaic cells

Lead–Acid Battery

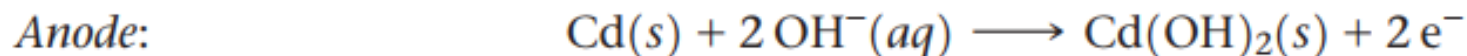
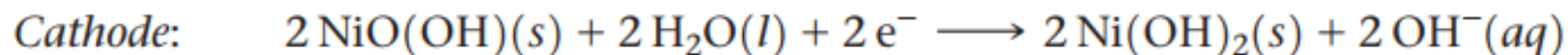


$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ}(\text{cathode}) - E_{\text{red}}^{\circ}(\text{anode}) = (+1.69 \text{ V}) - (-0.36 \text{ V}) = +2.05 \text{ V}$$

Advantage: rechargeable



Nickel Cadmium Battery

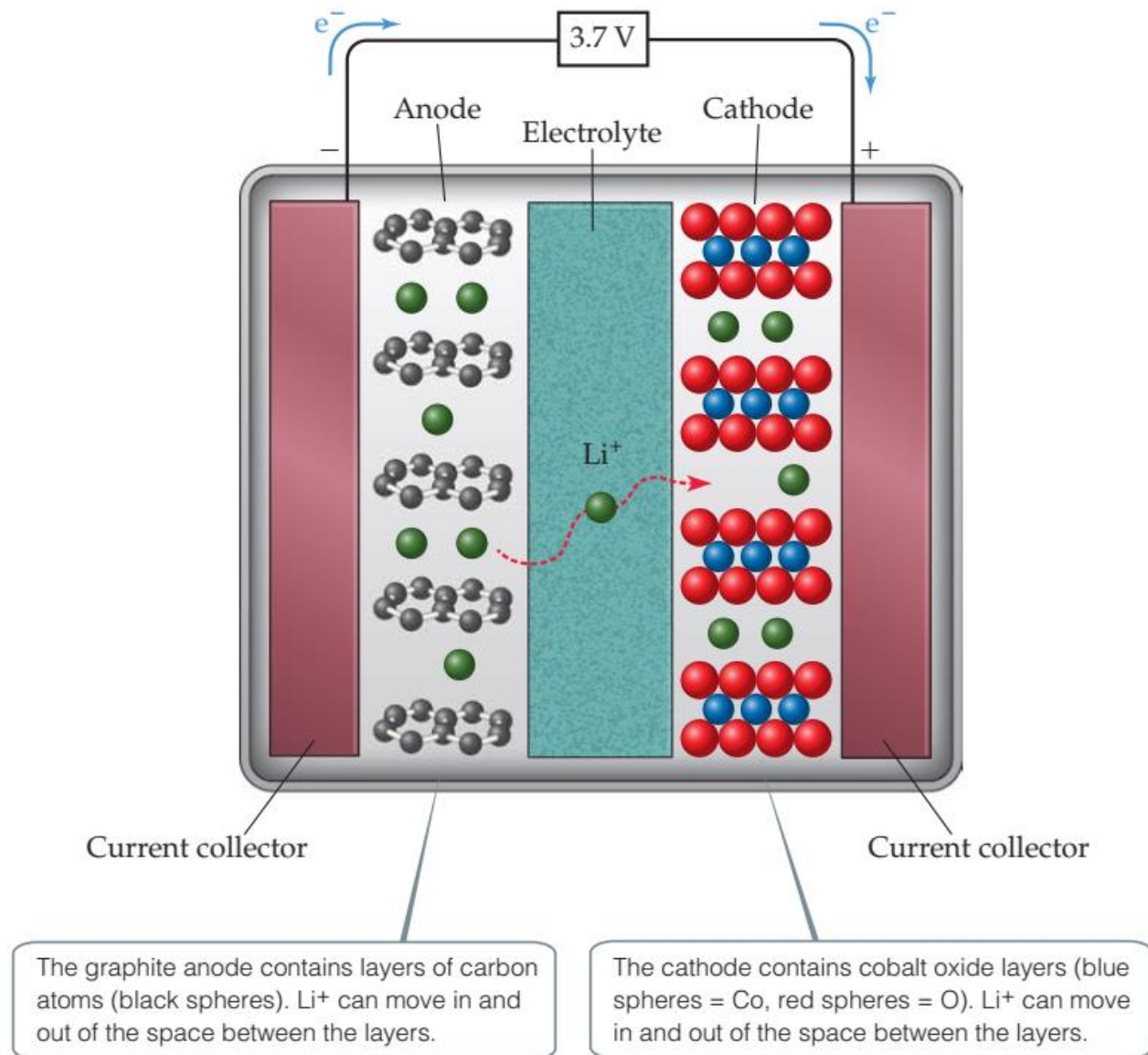


Advantage: rechargeable,

Disadvantage: cadmium is toxic

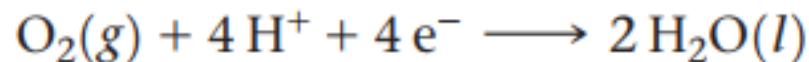
Application: portable electronic devices

Lithium Ion Battery



Fuel cell

Cathode:



Anode:



Overall:

